

Rajdeep Gill

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Education

University of Manitoba

Bachelor of Science in Computer Engineering

May 2026
4.46/4.5 GPA

Skills

Programming Languages: TypeScript, JavaScript, Python, C++, Java

Frameworks and Libraries: React, NextJS, PyTorch, OpenMPI, Node.js, Express, Flask

Tools: Git, Docker, PostgreSQL, Github Actions, Bazel, gRPC, Protocol Buffers (Protobuf)

Experience

Robinhood

Software Engineer Intern

May 2025 - Sept 2025

Toronto, ON

- Developed and shipped a real-time margin maintenance table built with React and Typescript, featuring embedded trade execution and risk indicators.
- Overhauled the margin experience by expanding the server-driven UI to include futures traders and unifying business logic across Web, iOS, and Android, resulting in a 43% increase in margin usage.
- Redesigned the margin call experience to include the maintenance table and support short selling functionality
- Enhanced a Protobuf-based server-driven UI architecture by implementing 10+ foundational components in Python and Typescript along with a logging framework to increase system usage and maintainability
- Repaired the core integration suite for the margin experience and added 20+ test suites, achieving 93% code coverage

Biosensors Research Lab

Machine Learning Research Intern

May 2024 - Dec 2024

Winnipeg, MB

- Created a custom convolutional neural network to predict cell viability from optical microscopy images, achieving 85% accuracy, establishing a non-invasive alternative to traditional staining methods for assessing cell health
- Developed a model validation GUI in Python displaying SHAP-based feature attributions, and visualizing cell images to debug model predictions
- Created a parallelized data processing pipeline with a new morphological parameter extraction algorithm, improving accuracy by 17% and cutting processing runtime by 75%.
- Replaced a high-risk system of manually-shared hard drives with a centralized MySQL database, allowing for cross-experiment queries and easy data sharing

Biosensors Research Lab

Machine Learning Research Intern

May 2023 - Aug 2023

Winnipeg, MB

- Built a framework for electronic testing equipment, allowing for automation of entire experimental sequences through Python scripts and cutting manual setup time
- Refactored legacy Python scripts to slash data analysis runtime by 90%, replacing inefficient loops with vectorized NumPy operations
- Developed a classification system using a Gaussian Mixture Model to determine cell viability from time-series impedance data, creating a probabilistic tool for assessing sample health on an unsupervised basis
- Programmed an FPGA to coordinate microfluidic device operations, integrating di-electrophoresis control with synchronized LED illumination and cell imaging

Projects

Void - AI App Builder (Next.js, TypeScript, Prisma, Docker)

- Developed an app to generate and deploy web apps from user prompts, integrating a usage-based Stripe billing system to monetize user compute time and API token consumption
- Implemented a persistent background job queue to process long-running AI build tasks asynchronously, allowing users to submit jobs and receive the completed result at a later time, even after closing the browser
- Utilized Docker for user code sandboxing, secure code execution and real-time application previews

Collaborative Document Editor (React, TypeScript, Socket.io, Flask)

- Engineered a collaborative text editor for up to 20 simultaneous users by broadcasting text operations, cursor locations, and online statuses, via WebSockets, achieving a sub-250ms synchronization latency
- Integrated Gemini to provide real-time suggestions for text, formatting and content generation, reducing writing time

UM-Tracker (React, TypeScript, PostgreSQL)

- Created a course assignment tracker for the University of Manitoba, allowing students to link their current courses to keep track of their upcoming assignments and exams
- Implemented syncing with personal calendars to display a comprehensive view of the user's schedule